

Mohammad Mostafavi – Senior Applied AI Research Scientist

AI Research Leader bridging computational pathology, event-based vision, and startup innovation.

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Interests

- Computational Pathology (Cell Detection, Tissue Segmentation, Sub-cell Models).
- Event-based (Neuromorphic) Cameras/Vision (Reconstruction, Super-resolution, Depth/Flow estimation).
- Super-resolution (RNNs for Event-cameras, weighted patches for Facial images).
- Generative Models (Diffusion-based, Solvers, Bridging, CTMs, Flow matching, and GANs).

Recent Experiences

- **Seoul National University – MPR lab- PostDoc** (2024.03~cont.).
 - InnoCORE (Innovation + Core Researchers) program (2025.09~2029.12)
Full time – hosted by {[KAIST-VI](#) (2025.09~2025.12), [DGIST-CVL](#) (2026.01~cont.)}
 - Machine Perception & Reasoning lab (2024.03~cont.), Automation and Systems Research Institute (ASRI) (2025.03~cont.).
 - Image to image Translation with Diffusion Models | Depth estimation with Event Cameras using Spiking Neural Networks | Versatile Incremental Learning | Mentored MS students | 2 Published papers [C1~C2] | 3 ongoing submissions.
 - <https://github.com/snumprlab/dbmsolver> ★ 8 [C1], <https://github.com/snumprlab/falcon> ★ 2 [C2] (2026.09).
 - **MIU-MISP** – Visiting PostDoc Researcher (2024.06~2025.02).
 - Medical Image and Signal Processing Center – Medical University of Isfahan
 - Semi-automated data labeling systems, deep learning models, and analysis software. | Conducting AI courses for university professors and medical experts. | Designed university curricula for AI in Ophthalmology and teaching it.
- **Lunit Inc. – Research Scientist (Senior)** (2021.06~2024.03) and **Team Leader** (2023.03~2024.03).
 - Led 3–4 researchers to deliver oncology AI models in clinical workflows, influencing 4 of 8 product development cycles.
 - Performance improvement for the main products (SCOPE) and pharma-requested models/bakeoffs.
 - Deep-learning-based cell detection and tissue segmentation models for SCOPE ([IO](#), [PD-L1](#)), and UIHC.
 - Establishing, persuading, and developing further product-oriented research directions.
 - Universal IHC models [A1, J6], Sub-cellular models [P1,P2], WSI synthesis, and End-point mutations.
 - Sharing recent CV/ML trends through study groups and weekly research seminars.
 - Publishing the findings, practices, and resources in Computational Pathology and PathOmics.
 - Abstracts [A1~A4], Journals [J3~ J6], Patents [P1,P2], and Challenges [S5].
 - Collaborating across pathology, biomedical engineering, product engineering, and business development.

Education

- **Ph.D. – Gwangju Institute of Science and Technology ([GIST](#)) – South Korea, Gwangju** (2015.09~2021.06).
 - Ph.D. in electrical engineering and computer science (Presidential Excellence Award).
 - Doctoral [Dissertation](#): Event-based vision: Image reconstruction, Super-resolution, Depth estimation.
 - Advisors: [Prof. Jonghyun Choi](#) ([GIST](#)), and [Prof. Kuk-Jin Yoon](#) ([KAIST](#)).
 - Top-tier journals (TPAMI/IJCV [J1, J2]), conferences (CVPR/IJCV [C3~C6] – 1 oral CVPR [C5]), and a Patent [P3] on Event-based vision for image reconstruction [J1, C5], super-resolution [J2, C4], and depth estimation [C3, C4].
 - Mentored MS students through paper submissions and cross-lab collaborations [C3].
 - Reaching **Rank #1** in the [CVPRW Event-based Vision](#) Competition (2021).
 - Open-sourced highly referred GitHub repos (from papers):
 - <https://github.com/gistvision/e2sri> ★ 51 [C5] (2026.06).
 - <https://github.com/yonseivnl/se-cff> ★ 52 [C3] (2026.06).

■ M.S. – Hakim Sabzevari University – Iran, Sabzevar.

■ M.Sc. in Electrical Engineering and Electronics.

■ Thesis: Facial Image **Super-Resolution** Using Weighted Patch Pairs.

■ Advisors: [Prof. Javad Haddadnia](#) (Hakim Sabzevari Uni.), and [Prof. Payman Moallem](#) (Uni. of Isfahan).

Selected Publications -15+ peer-reviewed publications (CVPR, TPAMI, IJCV, Nature npj Precision Oncology), 3 patents, and 780+ citations (h-index 10). [Google Scholar NNvELCcAAAAJ](#) | [ORCID 0000-0002-5883-3844](#) | [Scopus 40461500600](#)

Conferences *on Computer Vision / Machine Learning*

- [C1] “[DBMSolver: A Training-free Diffusion Bridge Sampler for High-Quality Image-to-Image Translation](#)”- S Venugopal, **M Mostafavi**, J Choi – CVF/IEEE – **CVPR 2026** (25.42% accept. rate) [\[Code\]](#)
- [C2] “[FALCON: Fast Adaptive Lightweight Computation of Intensities and Events for Depth Estimation](#)”- S Venugopal*, **M Mostafavi***, J Choi – *equal contribution – CVF/IEEE – **CVPR 2026 Findings** [\[Code\]](#)
- [C3] “[Stereo Depth from Event Cameras: Concentrate and Focus on the Future](#)”- Y Nam*, **M Mostafavi***, KJ Yoon, J Choi – *equal contribution – CVF/IEEE – **CVPR 2022** (25.33% accept. rate) [\[Code\]](#)
- [C4] “[Event-Intensity Stereo: Estimating Depth by the Best of Both Worlds](#)”- **M Mostafavi**, KJ Yoon, J Choi – CVF/IEEE – **ICCV 2021** (25.9% accept. rate)
- [C5] 🏆 “[Learning to Super Resolve Intensity Images from Events](#)” – **M Mostafavi**, J Choi, KJ Yoon – CVF/IEEE – **CVPR 2020** (5% accept. rate) [\[Oral\]](#)[\[Code\]](#)
- [C6] “[Event-based high dynamic range image and very high frame rate video generation using conditional generative adversarial networks](#)”- L Wang*, **M Mostafavi***, YS Ho, and KJ Yoon – CVF/IEEE – **CVPR 2019** (25.2% accept. rate)

Journals *on Computer Vision / Machine Learning*

- [J1] 🏆 “[E2SRI: Learning to Super-Resolve Intensity Images from Events](#)” – **M Mostafavi**, Y Nam, J Choi, KJ Yoon – IEEE-Transactions on Pattern Analysis and Machine Intelligence – **TPAMI 2021** (IF 24.31)
- [J2] “[Learning to reconstruct HDR images from events, with applications to depth and flow](#)” – **M Mostafavi**, L Wang, KJ Yoon – Springer- International Journal of Computer Vision – **IJCV 2021** (IF 11.54)

Journals *on AI-assisted oncology and Computational Pathology*

- [J3] “[Artificial intelligence-powered spatial analysis of tumor-infiltrating lymphocytes as a predictive biomarker for axitinib in adenoid cystic carcinoma](#)” – DH Kim, Y Lim, C-Y Ock, G Park, S Park, H Song, M Ma, **M Mostafavi**, EJ Kang, M-J Ahn, K-W Lee, JH Kwon, Y Yang, YH Choi, MK Kim, JH Ji, T Yun, S-B Kim, B Keam- **Head & Neck 2023** (IF 2.9)
- [J4] “[Artificial intelligence-powered whole-slide image analyzer reveals a distinctive distribution of tumor-infiltrating lymphocytes in neuroendocrine neoplasms](#)”- HG Cho, SI Cho, S Choi, W Jung, J Shin, G Park, J Moon, M Ma, H Song, **M Mostafavi**, M Kang, S Pereira, K Paeng, D Yoo, CY Ock, S Kim. **MDPI Diagnostics 2022** (IF 3.99)
- [J5] “[Changes in the tumor microenvironment in recurrent head and neck squamous cell carcinoma and its implication on efficacy of immune checkpoint inhibitors](#)” – DH Kim, M Kang, G Park, **M Mostafavi**, Y Lim, CY Ock, J Koh, YK Jeon, KC Jung, SH Ahn, EJ Chung, SK Kwon, B Keam – **Springer – Discover Oncology 2024**– (IF 2.8)
- [J6] “[A universal immunohistochemistry analyzer for generalizing AI-driven assessment of immunohistochemistry across immunostains and cancer types](#)” – B Brattoli*, **M Mostafavi***, T Lee*, W Jung, J Ryu, S Park, J Park, S Pereira, S Shin, S Choi, H Kim, D Yoo, SM Ali, K Paeng, CY Ock, SI Cho, S Kim- **Nature – npj precision oncology 2024** (IF 6.8)

Abstracts *on AI-assisted oncology and Computational Pathology*

- [A1] 🏆 “[Universal immunohistochemistry positivity classification of cancer cells across multiple cancer types and antibodies using artificial intelligence](#)” – B Brattoli*, **M Mostafavi***, S Choi, T Lee, S Kim, W Jung, SI Cho, J Lee, K Chung, J Ryu, S Park, S Pereira, S Shin, CY Ock – **AACR Annual Meeting Abstracts 2023**

- [A2] “1293 Fragmented pattern of tumor mass is related to fibroblast activation mitigating spatial interaction between tumor and immune cells” – S Kim, S Song, S Kim, M Kang, **M Mostafavi**, D Yoo, CH Ahn, S Ali, C-Y Ock- SITC Meeting Abstracts 2023
- [A3] “123P Artificial intelligence (AI)-powered analysis of human epidermal growth factor receptor-2 (HER2) and tumor-infiltrating lymphocytes (TILs) in advanced biliary tract cancer (BTC)” – G Kim, C Kim, B Kang, S Shin, T Lee, S Song, S Kim, **M Mostafavi**, H Song, S Pereira, H Chon- ESMO Congress Abstracts 2023
- [A4] “Performance validation of an artificial intelligence-powered PD-L1 combined positive score analyzer in six cancer types” – T Lee, SI Cho, S Choi, S Kim, W Jung, D Lee, S Lee, **M Mostafavi**, S Park, J Lee, J Shin, S Kim, K Paeng, CY Ock- ASCO Annual Meeting Abstracts 2023

Patents

- [P1] A method and apparatus for analyzing pathological slide images -2024.12- J Ryu., **M Mostafavi**, B Brattoli, CH Ahn, Y Lee, T Lee, S Kim, W Hwang, [US patent 2025](#), Korean (2024-0193383) {병리 슬라이드 이미지를 분석하는 방법 및 장치}.
- [P2] A method and apparatus for analyzing IHC stained images using an AI model -2024.02- B Brattoli, **M Mostafavi**, Y Lee, CH Ahn, T Lee, and J Ryu. Korean Patent (2024-0018904) {AI 모델을 이용한 IHC 염색 슬라이드 이미지 분석 방법 및 장치}.
- [P3] A method and apparatus for generating super resolve intensity image - 2020 - J Choi, **SM Mostafavi I**, and KJ Yoon. [Korean Patent \(102366187\)](#) {고해상도 강도 이미지 생성 방법 및 장치}.

Honors and Awards

- 🏆 Presidential Excellence Award – Best Ph.D. Dissertation – GIST (2021).
- 🏆 Rank #1 [CVPRW Event-based vision](#) competition for depth estimation from event cameras (2021).
- Outstanding RA Award – GIST (2020).
- Doctoral Consortiums: IEEE CVPR (2020 – USA, Virtual) and KCCV (2020 – Korea).
- Best paper awards: KSC (2019 – Korea), IPIU Bronze (2019 – Korea).
- Scholarships: Korean Gov. (2015-2019), Global Uni. Project (2015), Iranian Gov. Scholarship (2009-2011).

Languages

- English: Bilingual fluency, Farsi: Native, Korean: Low-Intermediate.

Programming Skills and Tools

- Programming Languages: Python, MATLAB, C++, Bash. ■ Libs. : PyTorch, OpenCV, TensorFlow, Keras.
- Tools: GNU Linux, Google GCP, CUDA (GPU), Docker, ROS, Git, Meshlab, LaTeX, Confluence, Jira, Notion.

Services

- [S1] Reviewer of CVPR, ECCV, ICCV, ICML, ICRA | MICCAI | Springer: IJCV | IEEE: MIP, TIM, TCI | IET: IP.
- [S2] Gold Reviewer – ICML 2026 – Complimentary Conference Registration.
- [S3] Lead Organizer – The ICML Chess Club: Blitz, Bullet, and Over-the-Board Networking – ICML 2026 Social.
- [S4] Program Committee – [Medical Imaging Analysis for Spatial Omics \(MISO\)](#) – MICCAI 2024, and 2026.
- [S5] Challenge organizer – [Advances in Neuromorphic Vision](#) – ICME 2024.
- [S6] Volume Editor – [MICCAI 2023 satellite events proceedings](#) – Springer LNCS 2024 .
- [S7] Challenge organizer – [OCELOT 2023: Cell Detection from Cell-Tissue Interaction](#) – MICCAI 2023.
- [S8] First Manager of [IEEE Young Prof. Affinity Group](#) in Gwangju Korea (2016).

Management Skills

■ Strategic Vision & Execution | Team & Organizational Leadership | Innovation & Change Management | Fundraising & Partnerships | Performance & Roadmap Management ■ Lunit Research/Tech Interview Committee (2021–2024) | Directed cross-functional teams delivering oncology AI models and shaping product strategy ■ OASIS Program (Korea): Startup creation, commercialization, and entrepreneurship (Modules 1, 2, 4, 5) | Stakeholder engagement across academia and industry.

Teaching

- Designed and taught AI curricula for medical practitioners (2024~Contd.)
 - ophthalmology, pathology, cardiology, dentistry, pharmacy
- Teaching Assistant – GIST- Korea (Spring 2020) Visual Recognition and Reasoning.
- Teaching Assistant – GIST- Korea (Spring 2017) Digital Signal Processing.
- Lecturer – Azad University – Iran (Fall 2011 ~ Spring 2013) Electronic circuits, and 6 labs.

References

Available on request.